

Name: Colin MurrayStudent Number: 101272354

Instructions: Your response must be typed in; handwritten responses will not be graded. Use a 12-point font size, with single line spacing.

Question Write an essay using 1400-1500 words using proper citations about the different ethical theories that were presented to you in the class lectures and the foundational framework for ethical deliberation.

Answer:

Ethical Theories and the Foundational Framework for Ethical Deliberation

Engineering is a profession that directly influences public safety, welfare, and quality of life. Engineers regularly make decisions that have wide ranging effects to whole communities, meaning requiring ethical deliberation is an essential part of professional practice. Ethical problems are rarely solved through technical knowledge alone; a framework is required for reasoning and eventual determination the best possible solution even if competing values, consequences or obligations are present. Throughout history several main ethical theories have been developed to help provide guidance for moral decision making.

1. Virtue-based Ethics
2. Rights-based Ethics
3. Duty-based Ethics
4. Consequence-based Ethics

[1]

While each theory offers valuable insight, no single theory on its own is sufficient. A complete approach to ethical decision making requires a comprehensive framework for ethical deliberation, this guides individuals through the process of understanding situations, evaluating options, and acting/making decisions responsibly. Under this lens, the four major ethical theories are examined and the relationship of this foundational framework for ethical reasoning and eventual practical resolution.

Ethics can be broadly understood as the study and practice of figuring out what is right vs wrong, these ideals then can be presented onto potential situations for deliberation and determination on what is the best course of action.

According to the lecture material, "Ethics is concerned with knowing and doing action that one believes will create positive results in one's life and in the lives of others." [1] Stating that any action is not neutral but an ethical issue that has consequences for oneself and others. Therefore, ethics is fundamentally concerned with identifying and promoting what is good and just while avoiding harm.

The first main theory to be discussed is virtue-based ethics, which originates from the teachings of the Greek philosopher Aristotle. Aristotle's ethical framework is teleological in nature, focusing on the goal or ultimate purpose (telos) towards which things are oriented. [1] In his view, ethical behaviour involves promoting the proper ends associated with human flourishing. Rather than concentrating primarily on rules or individual outcomes, virtue-based ethics emphasizes moral development.

Within that development is a series of actions which require the cultivation of virtues such as courage, honesty, justice, wisdom, and generosity. [1] Aristotle argued that these virtues are developed through consistent and constant practice by habit to eventually become stable characteristics of an individual's character. A truly virtuous person always acts in ways that reflect good judgment and moral standing. This judgment must involve an appropriate amount virtuous depending on the situation, finding an intermediate between excess and deficiency. [1] This "Golden Mean" is central to Aristotle philosophy, explaining how for example, courage lies between recklessness and cowardice, or generosity is between wastefulness and abundance.

For engineers, virtue-based ethics emphasizes the importance of developing professional judgment, integrity, and responsibility. Engineering competence is not only acquired through formal education but also through experience of constant and consistent cultivation of good habits.

However, virtue-based ethics has limitations, determining if someone possesses good character can be difficult, particularly when dealing with unknown people. Furthermore, individuals with positive intentions may still make poor decisions or cause harm in spite of possessing admirable character traits. [1] Consequently, virtue-based ethics provide valuable guidance but cannot be the sole basis for ethical deliberation.

Rights-based ethics is the second theory, which is principally associated with the work of the English philosopher John Locke and Genevan philosopher Jean-Jacques Rousseau. Rights-based ethics as a concept emerged in response to authoritarian political systems which lead to social inequalities, resulting in a rise of individual liberty. Locke argued that humans have natural rights because they are rational and capable of making their own decisions.

These are inalienable rights related to:

1. Life – Everyone has a fundamental right and duty to protect their own life, though a person can forfeit that protection by violating another's right to life through murder.
2. Liberty – People should be free to live and make choices as they please, so long as their actions don't harm or restrict the freedom of those around them.
3. Property – By mixing one's own labour with natural resources, you transform the common resource into private property – provided enough is left for others and nothing is let to waste through abundance/spoilage.

[2]

Rights-based ethics are particularly relevant for engineers when considering issues such as privacy, informed consent, workplace safety, intellectual property, and environmental effects. Especially in the modern age, engineers must ensure that technologies and systems do not unjustly infringe on the rights of individuals or communities.

Alike solely using virtue-based ethics, only considering rights-based ethics has its limitations. Too much emphasis on individual rights can encourage individualism at the detriment of the collective well-being of whole communities. [1] Additionally, conflicts can frequently arise when this individual thinking leads to competing groups while resulting in potential inequality for the less powerful.

The third major ethical theory is duty-based ethics, also known as deontological ethics, developed by German philosopher Immanuel Kant. Unlike virtue-based ethics, which focuses on character or rights-based ethics, which is related to liberty, duty-based ethics emphasizes moral obligations and universal principles. [1] The term “deontology” originates from the Greek word *deon*, meaning duty or obligation.

Kant thought that moral action are determined by the motives behind them rather than their consequences. He argued, individuals have a duty to act according to principles that could be universally applied to all people. [1]

This idea is expressed through Kant’s categorical imperatives:

1. Universal Applicability – “Always act according to a maxim that you would desire would be a universal law.”
2. Respect of Others - “Never act in such a way as to treat humanity, whether in ourselves or in others, as a means only but always as an end in itself.” [3]

This requires individuals to act only according to principles they would want everyone else to follow and to treat all with dignity. Therefore, from a duty-based perspective, ethical decision making involves asking if an action could be universally accepted while respecting the dignity of everyone.

In engineering, this approach can support professional accountability, honesty, fairness, and justice/adherence to professional codes of ethics. Engineers have a duty to protect and take into account public safety by maintaining this approach through reporting hazards, upholding professional competence regardless of personal gain, and more.

Despite its strengths, duty-based ethics also faces criticism. Ethical obligations can occasionally conflict with one another, creating unclear situations where it can be confusing what to prioritize in what order. Moreover, strictly following rules may fail to account for important contextual factors or human relationships. [1] Good intentions do not always lead to good outcomes, and a rigid application of universal rules may produce impractical results.

The fourth and final theory is consequence-based ethics, commonly known as utilitarianism, associated with English philosopher John Stuart Mill. Utilitarian ethics evaluates actions based on their outcomes and seeks to maximise overall happiness or benefit. Ethical decisions in this framework should promote the greatest good for the greatest number of people. [1] Under utilitarianism, the moral value of an action depends entirely on its consequences. Decision makers must weigh potential costs and benefits to select the course of action that will be the most favourable overall outcome.

This approach is frequently reflected in engineering practice through cost-benefit analysis, risk assessments, and policy evaluations. Utilitarian reasoning is particularly useful when large scale projects affect many stakeholders. Engineers often evaluate different options based on factors such as safety, efficiency, economic benefits, and environmental impacts.

Utilitarianism also has significant limitations like the other three theories. Determining what constitutes as the “greatest good” can be subjective and falls on opinion with the absence of fact. Furthermore, the interests of minorities or vulnerable groups/individuals may be sacrificed in order to benefit the overall or a larger group or individual. It can also be argued certain actions remain morally wrong regardless of any beneficial consequences that may be produced as a result.

Although each ethical theory offers valuable insights, the lecture material and logical reasoning both conclude that no single theory is sufficient for addressing all ethical problems. Real world ethical dilemmas are often complex, with considerations of character, rights, duty, and consequences simultaneously. As a result, ethical theories should be viewed as sources of ethical insight rather than complete solutions.

A full foundational framework would not focus on individual ethical theories but the underlying process through which humans come to know what is true and determine what is good. [1] This is done through a process of four stages:

1. Experience – Serves as the ethical foundation for all subsequent ethical reasoning by providing information and a point of contact with the world.
2. Understanding – Interpret prior experiences by asking questions, exploring possibilities, identifying relationships, and thinking about explanations for events.
3. Judging – Once a possible interpretation is developed, the accuracy of it is determined with reasoning.
4. Decision – Ethical reasoning ultimately requires moving from prior knowledge and experience to action. A decision on what is to be done is determined and action and then completed accordingly.

[1]

In conclusion, ethical decision making is an essential component of engineering practice for its role in choices that affect public safety, welfare, and quality of life. No single theory can explain the best way to make those choices, but knowledge and understanding of the basis of these theories can help form a framework. Virtue-based ethics emphasizes character and moral development, rights-based ethics focuses on protecting individual freedoms, duty-based ethics highlights universal obligations and moral principles, and consequence-based ethics evaluates the outcomes of actions. The foundational framework for ethical deliberation builds on these theories by guiding individuals through the processes of experience, understanding, judgment, and decision making. By drawing upon both ethical theories and a structured deliberative process, engineers can make responsible decisions that promote the public good, uphold professional integrity, and contribute positively to society.

References:

- [1] C. Jamieson, Ethics 101, ECOR 4995A Professional Practice Lecture 9, Carleton University, Ottawa, ON, Canada, 2025. [Online]. Available: <https://brightspace.carleton.ca/d2l/le/content/420552/viewContent/4688218/View> [Accessed: Jun. 3, 2026].
- [2] M. H. Lessnoff, "Locke: Natural Rights to Life, Liberty, and Property," Political Science Institute. [Online]. Available: <https://polsci.institute/classical-political-philosophy/locke-natural-rights-life-liberty-property/>. [Accessed: Jun. 3, 2026].
- [3] I. Kant, Foundations of the Metaphysics of Morals, trans. L. W. Beck. New York, NY, USA: Library of Liberal Arts, 1959, p. 421.